

DevOps & Cloud Deployment – Module 3 Exercise

Hands-on Exercise No. 3

Course: DevOps & Cloud Deployment

Module: Containerization with Docker

Total Marks: 10

Due Date: 17/06/2026

Instructions

Prepare your solution in MS Word.

Submit in .doc or .docx format.

Late or copied submissions will not be accepted.

Learning Outcome

After completing this exercise, you will be able to:

- Understand Docker architecture and container concepts.
 - Build Docker images for applications.
 - Run and manage containers.
 - Use Docker Compose for multi-container applications.
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Problem Statement

You are working as a DevOps Engineer for a SaaS company. The client wants the application to be containerized so it can run consistently across environments. Your task is to build Docker images and run containers for a Node.js application.

Task Requirements

Create a simple Node.js application (`server.js`) that returns:
“**Hello from Docker!**” on `localhost:3000`.

Write a `Dockerfile` with the following steps:

- Use official Node.js base image.
- Copy project files into the container.

- Install dependencies using `npm install`.
- Expose port 3000.
- Start the application using `node server.js`.

Build the Docker image using:

```
docker build -t myapp .
```

Run the container using:

```
docker run -p 3000:3000 myapp
```

Verify the application in the browser at `http://localhost:3000`.

Create a `docker-compose.yml` file to run the Node.js application along with a MongoDB service.

Take Screenshots of

- Dockerfile code.
 - Image build process in terminal.
 - Running container output in browser.
 - Docker Compose configuration and execution.
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Submission Guidelines

Upload your `.docx` file with Dockerfile, `docker-compose.yml`, screenshots, and explanations via LMS.